

1. (15 points) An insulated cylinder contains 5.0 liters of air at room temperature (300 K) and atmospheric pressure. (10^5 N/m^2). At one end of the cylinder is a massless piston, whose surface area is 0.01 m^2 . Suppose that you push the piston in very hard, exerting a force of 2500 N. The piston moves 4 mm before it is stopped.

a. (5 points) Calculate the amount of work you did on the system, and the heat added to the gas. Assuming that all of the energy added goes into the gas (not the piston or cylinder walls), by how much does the internal energy of the gas increase?

$$W = f \cdot d$$

$$W = 2500(0.04) = 100 \text{ J}$$

↑
units

$$-\Delta F = W$$

increases by 100 J
 $\Delta u?$

~~int~~

$Q?$

b. (5 points) Calculate the change in the entropy of the gas once it has again reached equilibrium. (Because the piston displacement is tiny compared with the total volume of gas, you may assume the internal pressure and temperature of the gas are constant in this problem).

in equilib.

$$\Delta S = \frac{\Delta H - \Delta G}{T}$$

$$\Delta S = \frac{\Delta U}{T} + \frac{P\Delta V}{T}$$

$$Q = T\Delta S$$

$$\Delta S = \frac{\Delta U}{T} = \frac{100}{300} = \frac{1}{3} \text{ J/K}$$

$$Q = NkT \ln \frac{V_f}{V_i} \approx \ln 0$$

$$Q = 0$$

$$\Delta U = Q + W = 0 + 100 = 100$$

c. (5 points) Is this process reversible or irreversible? Explain. If it is reversible, how could you make it irreversible? Likewise, if it is irreversible, how could you make it reversible?

Irreversible b/c $\Delta S < \frac{Q}{T}$; there is a gain in entropy when the system is returned

To make this reversible the piston needed to

2. (15 points) The multiplicity for a cold, large Einstein solid is:

$$\Omega \approx \left(\frac{eN}{q} \right)^q$$

With the system's total energy defined by:

$$U = \epsilon q$$

$$U = 3NkT$$

(a) (5 points) Calculate the temperature of the solid as a function of its total energy.

$$dU = TdS - PdV + \mu dN$$

$$\partial S = \frac{dU + PdV - \mu dN}{T}$$

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U} \right)_{N,V}$$

$$\frac{1}{T} = \left(\frac{\partial U + PdV - \mu dN}{\partial U} \right) \Rightarrow 1 = PdV - \mu dN$$

$$S = k \ln \Omega = 8.62 \times 10^{-5} \frac{\text{eV}}{\text{K}} \ln \left(\frac{eN}{q} \right)^q$$

$$\ln(N!) \approx N \ln N - N$$

$$= 8.62 \times 10^{-5} \ln q \cdot \ln eN - \ln q \ln q$$

$$= 8.62 \times 10^{-5} q \ln N$$

(b) (5 points) Calculate the energy of the solid as a function of temperature.

$$dU = TdS - PdV + \mu dN$$

$$\partial U = \partial S \cdot T \quad \text{from} \quad \frac{1}{T} = \left(\frac{\partial S}{\partial U} \right)_{N,V}$$

$$\partial U = C_v \cdot \partial T$$

$$U = 3NkT$$

~~$$F = HF - HC$$~~

(c) (5 points) Calculate the constant volume heat capacity of the solid.

$$C_v = \left(\frac{\partial U}{\partial T} \right)_v \leftarrow \text{use that}$$

$$\partial U = T \partial S - P \partial V + \mu \partial N$$

$$U = 3NKT$$

3. (15 points) Calcium carbonate has two common crystalline forms, calcite and aragonite, with the following thermodynamic data:

Form	$\Delta_f H$ (kJ)	$\Delta_f G$ (kJ)	S (J/K)	V (cm ³)
calcite	-1206.9	-1128.8	92.9	36.93
aragonite	-1207.1	-1127.8	88.7	34.15

All data are for one mole of material at 298 K and 1 bar. S and V can be considered independent of temperature for this problem.

a. (5 points) Which form is stable at the earth's surface and why?

3 calcite b/c it has a higher entropy
 all things trend to increase entropy.
 Also, $\Delta_f H$ for aragonite is higher so it is

b. (5 points) Assume the temperature is kept at room temperature. Calculate the pressure at which the phase not stable in part a should become stable.

$$P = T \left(\frac{\partial S}{\partial V} \right)_{N,T}$$

$$P = 300 \left(\frac{88.7}{34.15} \right) \approx 7.8 \times 10^4 \text{ N/m}^2$$